

James R Rose

412-445-8323

Jrose835@gmail.com

307 Candler St NE

Atlanta GA, 30307

Computational biologist PhD with expertise in spatial transcriptomics, next generation sequencing, epigenetics, single cell data analysis, machine learning, and immunology/autoimmunity

Looking for opportunities to apply my quantitative & computational analysis skills towards interesting biological questions in translational research

Research Experience

Assistant Scientist, Department of Biomedical Engineering
Santangelo Lab Emory University, Atlanta GA

Sept 2024 to Present

- Built bioinformatic analysis pipeline for large spatial transcriptomic project using the 10x Xenium image-based technology assaying mRNA delivery across multiple tissues and species
- Developed tool to integrate spatial transcript data with immunofluorescence images to validate efficiency of RNA delivery and protein production in lung tissue
- Designed and automated NGS bioinformatic (RNAseq, single-cell, etc) pipelines and large data analysis for group using Nextflow pipelines

PhD Graduate Researcher, Emory Genetics and Molecular Biology program
Boss Lab Emory University, Atlanta GA

Aug 2019 to Aug 2024

- Analyzed complex multi-modal bulk and single-cell sequencing datasets (RNAseq, ATACseq) for characterizing disease clinical subtypes, immune cell differentiation, and gene expression regulation
- Performed genetic and epigenetic biomarker discovery for SLE and other autoimmune diseases

Senior Associate Scientist, Janssen R&D, Biotherapeutics Development
Johnson & Johnson, Malvern PA

Aug 2016 to Jun 2019

- Led a team of development researchers on drug process development, new technology implementation, and regulatory submissions for therapeutic product worth \$6B in annual sales
- Implemented statistical process control technologies for monoclonal antibody manufacturing at international GMP facilities

Engineer I, Cell Culture Technical Development
Biogen, Durham NC

Jun 2013 to Aug 2016

- Developed mammalian cell culture processes for monoclonal antibody production at large-scale manufacturing

Education

PhD Genetics & Molecular Biology, Emory University, 2019-2024

Master's Biotechnology, The Pennsylvania State University, 2011-2013

B.S. Biotechnology, The Pennsylvania State University, 2008-2013

Skills

Data Analysis

- Next generation sequencing (NGS) bioinformatics analysis & pipeline development (single-cell/bulk RNAseq, ATACseq, CITEseq)
- Nextflow pipeline automation
- Data visualization (ggplot, seaborn)
- Machine learning, statistical modeling
- Differential expression testing (DESeq2)
- Spatial transcriptomics (10x Xenium)
- Image analysis (QuPath, ImageJ, OpenCV)

Programming

- R (tidyverse), python, bash, groovy
- Shiny App development
- Git version control, Docker, Rstudio server
- Rmarkdown & jupyter notebooks
- Remote/cloud server operations, AWS

Domain Knowledge

- Immunology (B & T cell)
- Epigenetics

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Selected Publications

- Pramod Kumar EK, Younghun Jung, Jae Yeon Joo, Elex S. Rose, **James R. Rose**, et al. Hyperpotent polymer formulations for nucleic acid delivery to the respiratory tract (*In preparation*)
- **Rose J. R.**, Ishii Y., et al. Clinically inactive SLE resting naïve B cells retain distinct transcriptomes and alterations to the epigenome. (*In preparation*)
- **Rose, J. R.**, Akdogan-Ozdilek, B., et al. Distinct transcriptomic and epigenomic modalities underpin human memory T cell subsets and their activation potential. *Commun Biol* **6**, 363 (2023).

Relevant Projects

[SLE Single Cell Disease Atlas](#)

Jan 2024 to Aug 2024

- Utilized public single-cell sequencing data from multiple sources to compile, qc, and integrate reference dataset comprised of B cells from systemic lupus erythematosus (SLE) patients
- Using advanced AI-based data integration methods to query reference data with in-house generated data sets to derive biological insights into individual samples and disease endotypes
- Applying machine learning techniques to classify disease endotypes based on sequencing data

[Inspirit AI Learning Fellow](#)

Jul 2023 to Aug 2023

- Short course on AI and machine learning methods through the Inspirit program taught by experts in the industry (Google, Stanford, Meta)
- Completed python-based coding project exploring embeddings, natural learning processing (NLP), and multi-class prediction models to classify textual data

Teaching Experience

Course Lecturer & Teaching Assistant

Dec 2021 to Dec 2023

IBS519: Statistical rigor, reproducibility, and experimental design

- Developed teaching materials for revamped course curriculum on quantitative data analysis
- Lectured on [data visualization principles](#) and use of ggplot2 in R to incoming graduate students

Social Media Contacts



Github: <https://github.com/jrose835>



Personal Website: https://jrose835.quarto.pub/jimrose_quartosite/



Mastodon: <https://genomic.social/@jrose835>



Blue Sky: <https://bsky.app/profile/jrose835.bsky.social>